

DEV TECH SPEC

1. Core Functionality

1. Style Extraction

- Extract aesthetic from one reference photo:
 - Tone curve
 - Contrast behavior
 - Color grading / HSL characteristics
- Output a usable “style profile.”

2. Lighting-Aware Adaptation

- Analyze each batch image for lighting/exposure.
- Adjust the style so the look stays consistent across mixed lighting.
- Preserve skin tones and highlight/shadow detail.

3. Batch Processing

- Apply the style profile to all uploaded photos (20–50).
- Generate edited JPEGs.
- Produce quick previews (first 2–3 images).

4. XMP Preset Generation

- Convert the extracted style into a Lightroom-compatible `.xmp` preset.
- Package this with edited images.

5. ZIP Export

- Bundle:
 - Edited JPGs
 - XMP preset

- Return a downloadable ZIP file.
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2. User Flow (Simple)

1. Upload reference image.
 2. Upload batch images.
 3. System processes (style extraction → batch editing → XMP generation).
 4. User sees preview thumbnails.
 5. User downloads ZIP.
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3. Required Backend Components

1. Upload Handling

- Support multiple batch images (JPEG/PNG).
- Temporary storage (S3 or local).

2. Processing Worker / Service

- Runs style extraction model.
- Runs lighting-aware batch processing.
- Generates final edited images.
- Creates the XMP file.

3. Job Queue (Simple)

- Queue jobs (queued → processing → done).

4. ZIP Packaging

- Bundle results and return download link.
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4. Frontend Requirements

1. Drag-and-drop upload for:
 - Reference image
 - Batch images
 2. Simple status/progress indicator.
 3. Show 2–3 before/after previews.
 4. Download button for ZIP file.
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5. Non-Functional (MVP)

- Max batch size: 20–50 images.
- Processing target: <3 minutes per job (flexible).
- Basic error handling (bad file type, too many files).
- Auto-cleanup of temporary files.

BUDGET

Milestone Costs:

1. Style Extraction — **\$385**
2. Single Image Application — **\$245**
3. Batch Processing — **\$350**
4. XMP + ZIP Export — **\$315**
5. UI/UX — **\$245**
6. Optimization + QA — **\$210**

Total MVP Cost: \$1,750

Timeline: 2 months total